ALEXANDER SANCHEZ CYRIL

Software Engineer — Full-Stack, Backend & AI Systems

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PROFESSIONAL SUMMARY

Software engineer with 3+ years of experience shipping full-stack, backend, and AI-powered systems across independent projects and collaborative engineering teams. Author of ApiPilot and Portfolio.Resilience, production-grade .NET API-boundary and resilience libraries with real-world adoption, and current Open Source Engineering Fellow at ArcellAI, building provenance-aware AI systems for economic analysis. Skilled in scalable API design, authentication, distributed architectures, and agentic AI workflows across Python, TypeScript, C#, Java, and Go.

KEY ACHIEVEMENTS

- Built scalable APIs, authentication services, messaging systems, and database-driven applications in Python, TypeScript, C#, Java, and Go across Django, FastAPI, ASP.NET Core, Fastify, Gin, and Next.js
- Delivered SaaS platforms, AI-powered applications, and developer-productivity tools end to end, from architecture through deployment
- Applied distributed-systems patterns — caching, async messaging, event-driven workflows, and session management using Redis, RabbitMQ, and Kafka
- Deployed cloud-hosted applications with Docker, CI/CD pipelines, and Infrastructure as Code (Terraform) on Vercel, Render, and Neon
- Authored and published Portfolio.Resilience and ApiPilot on NuGet; built File-Ferry's SchemaLence workspace for PostgreSQL schema browsing, data previews, exports, backups, and cloud files downloads.

TECHNICAL SKILLS

Category	Skills
Languages	Python · JavaScript/TypeScript · C# · Java · Go
Backend & APIs	ASP.NET Core · Spring Boot · Django · FastAPI · Express · Fastify · Gin
Frontend	React · Next.js · Tailwind CSS · HTML5/CSS3
AI & Agentic Systems	LangGraph · LangChain · LangSmith · LlamaIndex · RAG · MCP · Vector Databases · Multi-Agent Systems
Databases	PostgreSQL · MongoDB · MySQL · SQLite
Cloud & DevOps	Docker · GitHub Actions (CI/CD) · Terraform · Vercel · Render · Neon
Distributed Systems	Microservices · Event-Driven Design · REST APIs · JWT Auth · RabbitMQ · Kafka · Redis
Tools	Git · Postman · Swagger/OpenAPI · Linux

PROFESSIONAL EXPERIENCE

Open Source Engineering Fellow — ArcellAI Inc.

Sept 2026 – Present

San Francisco, CA, USA (Remote) · arcell.ai

- Contribute to VEDA (Vendor Economic Dependency Assessment), a provenance-aware AI system for vendor economic dependency and related-party analysis
- Design entity-resolution pipelines, multi-source evidence retrieval, normalization, conflict detection, and human-review workflows centered on a relationship/evidence context graph
- Contribute to federated system architecture, source connectors, and agent design, including canonical modeling, semantic layer, and ontology mapping for context-driven agents

Self-Employed Software Engineer

April 2023 – Present

End-to-end owner of architecture, backend, frontend, database design, API development, cloud deployment, and maintenance across several independent projects, summarized below.

Portfolio-Resilience & ApiPilot — Open-Source .NET 10 Library

Author & Maintainer · NuGet: Portfolio-Resilience · GitHub: sancy1/portfolio-resilience · MIT License

- Designed and shipped two zero-third-party-runtime-dependency .NET 10 libraries on NuGet: Portfolio.Resilience for outbound-call protection and ApiPilot for contract-first ASP.NET Core API boundaries.
- Portfolio.Resilience: retry, circuit breaker, timeout, rate limiting, bulkhead, hedging, structured logging, latency metrics, correlation IDs, idempotency-key propagation, PCI-safe event scrubbing, and a Payment Processing Preset.
- ApiPilot: standardized JSON response/error envelopes, validation, pagination, correlation IDs, CSRF protection, secure cookies, Origin/Fetch Metadata policy, OpenAPI emitter, rate-limit integration, and a zero-dependency browser client.
- Reached 548 automated tests across Windows and Linux CI for Portfolio.Resilience; adopted in production by File-Ferry across 10,000+ directory walks.

EL-Exousia Weather Intel — Agentic Weather Intelligence Platform

Stack: FastAPI · React · LangGraph · LangChain · LangSmith · LlamaIndex · MCP

- Built an AI-powered weather intelligence system using multi-agent orchestration and tool-calling LLM workflows in LangGraph
- Designed a retrieval-augmented generation (RAG) pipeline with vector memory for contextual and historical weather reasoning
- Integrated real-time weather APIs into the agentic pipeline to drive dynamic forecast generation
- Implemented modular, multi-step agent workflows for structured decision-making and response generation

AxioQuan — E-Learning & Marketplace Platform

Stack: Java · C# · Fastify (TypeScript) · Next.js

- Designed and built a full-stack e-learning and digital marketplace platform with course creation, quizzes, and paid content delivery
- Implemented role-based access control for students, instructors, and administrators
- Built real-time messaging between students and instructors, plus a payment flow for enrollment and content access
- Engineered the backend APIs and database schema for scalable content management and delivery

LEADERSHIP & ACADEMIC PROJECT EXPERIENCE

- Led student project teams in university software development courses, coordinating planning, task distribution, and system design decisions
- Served as technical lead in Agile-style environments, managing sprint planning and feature delivery
- Supported the team through code reviews, debugging help, and technical guidance
- Applied core engineering practices — Git workflows, testing, and system design to deliver team projects from requirements to deployment

EDUCATION

Bachelor of Science in Software Development — Brigham Young University–Idaho *2026*

Associate of Applied Science in Software Development — Brigham Young University–Idaho *2026*

CERTIFICATIONS

- Software Development — Brigham Young University–Idaho, 2026
- Web Development — Brigham Young University–Idaho, 2025
- Web & Computer Programming — Brigham Young University–Idaho, 2024